

Solutions (Worksheet-6).

$$\begin{aligned}\textcircled{\text{I}} \quad \text{Cov}(X, Y) &:= E[(X - E[X])(Y - E[Y])] \\ &= E[(XY - XE[Y] - YE[X] + E[X]E[Y])] \\ &= E[XY] - E[XE[Y]] - E[YE[X]] + E[E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y] \quad (\because E[Y]E[X] = E[X]E[Y])\end{aligned}$$

$$\begin{aligned}\textcircled{\text{II}} \quad \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] \quad (\text{By } \textcircled{\text{I}}) \\ \therefore \text{Cov}(X, X) &= E[X^2] - (E[X])^2 = E[X^2] + (E[X])^2 - 2E[X] \cdot E[X] \\ &= E[(X - E[X])^2] = \text{Var}(X)\end{aligned}$$

$$\begin{aligned}\textcircled{\text{III}} \quad \text{Cov}(aX, Y) &= E[(aX)(Y)] - E[aX]E[Y] \quad (\text{By } \textcircled{\text{I}}) \\ &= aE[XY] - aE[X]E[Y] \\ &= a(E[XY] - E[X]E[Y]) = a \text{Cov}(X, Y)\end{aligned}$$

$$\begin{aligned}\textcircled{\text{IV}} \quad \text{Cov}(X+Z, Y) &= E[(X+Z)Y] - E[X+Z]E[Y] \quad (\text{By } \textcircled{\text{I}}) \\ &= E[XY + ZY] - (E[X] + E[Z])E[Y] \\ &= E[XY] + E[ZY] - E[X]E[Y] - E[Z]E[Y] \\ &= \text{Cov}(X, Y) + \text{Cov}(Z, Y)\end{aligned}$$